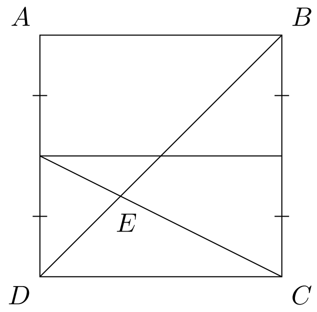
NC(SMC)^2 2026 Accuracy Round

NC(SMC)^2 Problem Writers

May 9th, 2026

1. What is the average of the first 10 positive even numbers?
2. A stick of length 2026 is broken into three pieces of integer length. What is the largest possible integer length of a side of a triangle formed by the three pieces?
3. Susan draws one card from a standard deck of 52 cards. Given that the probability that the card is a face card (Jack, Queen, or King) is $\frac{m}{n}$ in simplest form, what is $m+n$?
4. Square $ABCD$ with side length 60 is divided by several lines as shown. What is the area of triangle DEC?



5. Find the integer x which satisfies
$$\frac{1}{x}+\frac{1}{x+2}+\frac{1}{x+4}+\frac{1}{x+6}+\frac{1}{x+8}+\frac{1}{x+10}=0$$
6. The language of Globalese has 6 consonants and 4 vowels. A word in Globalese consists of either only vowels or only consonants, and letters cannot be repeated in a word. How many three letter words are there in Globalese?
7. The smallest prime number with a sum of digits equal to 13 is 67. Find the next smallest prime number with a sum of digits equal to 13.
8. A function f satisfies
$$3f(x)+2x\cdot f(1/x)=\frac{1}{x^2}$$
for all positive real x. The value of $f(2)$ can be represented as $-\frac{m}{n}$ in simplest form. Compute $m+n$.

9. Parallelogram $ABCD$ has $AB = 2$, $BC = 3$, and angle $ABC = 120°$. Point E lies on AD such that BE bisects angle ABC. The area of trapezoid $BCDE$ is $\sqrt{n}$. Compute n

10. Alan draws a hexagon (not necessarily regular) on a piece of paper. Aaron draws a heptagon (seven-sided polygon) on top of Alan's hexagon. Each side of the hexagon intersects any given side of the heptagon at at most one point. What is the maximum number of intersection points of the two drawings?